

Developer Guide (Part 5): Backtesting Framework Design & Strategy Development Lifecycle

Version: 1.0

Audience: Software Engineers, Quant Developers, Data Engineers, ML Engineers

Prerequisites:

- Part 1 – Introduction & System Overview
 - Part 2 – Financial Markets & Market Data Fundamentals
 - Part 3 – Mathematical Foundations
 - Part 4 – Data Engineering & Feature Engineering
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1. Purpose

Backtesting is the process of evaluating a trading strategy using historical market data. It answers a fundamental question:

"If this strategy had been used in the past, how would it have performed?"

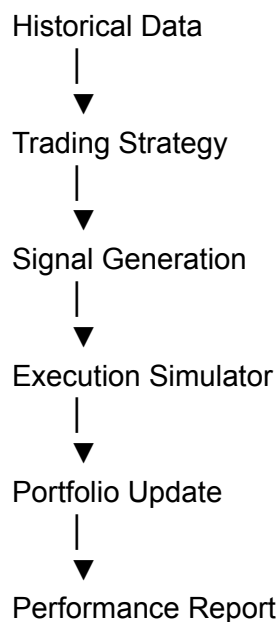
Backtesting is not intended to predict the future with certainty. Instead, it helps developers understand:

- Whether the strategy behaves as expected.
 - How much risk it carries.
 - Under which market conditions it succeeds or fails.
 - Whether it is robust enough for further testing.
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2. What is Backtesting?

A backtest simulates historical trading by applying strategy rules to historical market data.

The engine processes each historical event as though it were happening in real time.



Unlike live trading, a backtest does not interact with a broker or exchange. Instead, it uses simulated execution based on historical prices and predefined assumptions.

3. Why Backtesting Matters

Without backtesting:

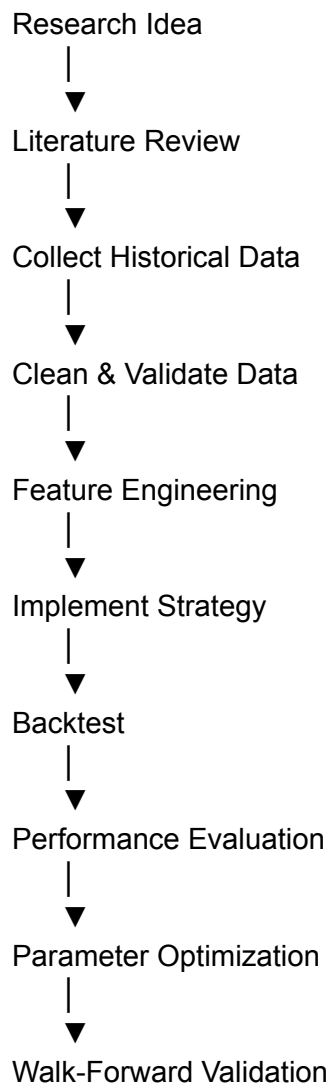
- Strategies cannot be validated.
- Risks remain unknown.
- Bugs may go unnoticed.
- Performance cannot be measured objectively.

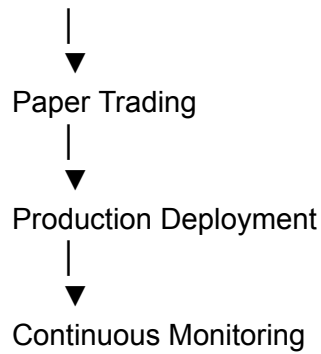
Backtesting helps developers:

- Compare multiple strategies.
 - Tune parameters responsibly.
 - Measure returns and risk.
 - Identify weaknesses before deployment.
 - Build confidence before paper or live trading.
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4. Strategy Development Lifecycle

A professional quant workflow follows a structured lifecycle.





Each stage builds on the previous one and should be repeatable and documented.

5. Backtesting Workflow

A typical backtesting process includes:

Step 1 – Load Historical Data

Examples:

- OHLCV data
 - Tick data
 - Order book snapshots
 - Corporate actions
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Step 2 – Generate Features

Examples:

- Moving averages
 - RSI
 - MACD
 - Rolling volatility
 - Z-score
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Step 3 – Generate Trading Signals

Example rules:

- Buy when 20-day SMA crosses above 50-day SMA.
- Sell when RSI exceeds 70.

Signals should be generated using only information available at that point in time.

Step 4 – Simulate Order Execution

The backtester simulates:

- Market orders
 - Limit orders
 - Partial fills (if modeled)
 - Execution delays
 - Slippage
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Step 5 – Update Portfolio

Track:

- Cash balance
 - Positions
 - Unrealized P&L
 - Realized P&L
 - Portfolio value
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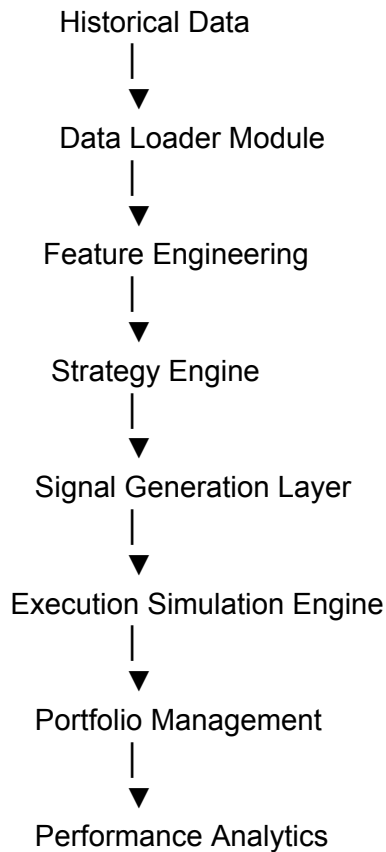
Step 6 – Generate Reports

Typical outputs:

- Equity curve
 - Drawdown chart
 - Trade log
 - Risk metrics
 - Performance summary
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6. Backtesting System Architecture

A modular architecture makes the framework easier to maintain and extend.



Responsibilities

Component	Responsibility
Data Loader	Reads historical market data
Feature Engine	Calculates indicators and derived features
Strategy Engine	Implements trading rules
Execution Engine	Simulates order execution
Portfolio Manager	Tracks positions and cash
Analytics Engine	Computes performance metrics

7. Core Components of a Backtesting Engine

Data Loader

Responsible for:

- Reading historical datasets
 - Handling multiple file formats
 - Time synchronization
 - Corporate action adjustments
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Strategy Engine

Encapsulates trading logic.

Responsibilities:

- Generate buy/sell signals
 - Manage strategy parameters
 - Prevent future data leakage
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Execution Engine

Simulates how orders would have been filled.

Supports:

- Market orders
 - Limit orders
 - Stop orders
 - Execution latency
 - Partial fills
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Portfolio Manager

Tracks:

- Available cash
 - Current holdings
 - Average entry price
 - Portfolio valuation
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Performance Analyzer

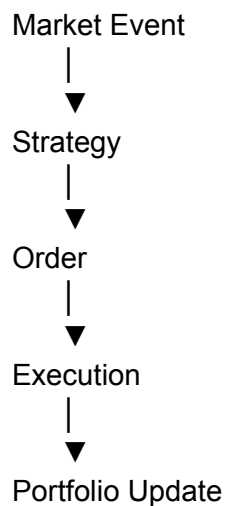
Produces:

- Returns
 - Drawdowns
 - Trade statistics
 - Risk metrics
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8. Event-Driven vs Vectorized Backtesting

Event-Driven

Processes one event at a time, similar to live trading.



Advantages

- Closely resembles production systems
- Easier to model realistic execution

Disadvantages

- Slower on large datasets
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Vectorized

Uses vectorized operations over entire datasets.

Advantages

- Faster
- Ideal for research and experimentation

Disadvantages

- Less realistic for execution modeling
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9. Order Execution Simulation

Execution assumptions significantly affect results.

Common execution models:

Market Order

Executes immediately at the current market price.

Limit Order

Executes only if the specified price is reached.

Stop Order

Activates after a trigger price is crossed.

Partial Fill

Only part of the requested quantity is executed due to limited market liquidity.

Realistic execution modeling improves the reliability of backtest results.

10. Transaction Costs & Slippage

Ignoring trading costs can make an unprofitable strategy appear profitable.

Transaction Costs

Include:

- Brokerage fees
 - Exchange fees
 - Taxes
 - Regulatory charges
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Slippage

Difference between the expected execution price and the actual execution price.

Causes

- Market volatility
- Large order sizes
- Limited liquidity

Strategies with frequent trading are particularly sensitive to these effects.

11. Performance Metrics

A backtesting engine should report multiple metrics rather than focusing solely on returns.

Metric	Description
Total Return	Overall profit or loss
CAGR	Annualized growth rate
Sharpe Ratio	Return adjusted for total volatility
Sortino Ratio	Return adjusted for downside volatility
Maximum Drawdown	Largest decline from peak portfolio value
Win Rate	Percentage of profitable trades

Profit Factor	Gross profit divided by gross loss
Average Trade	Mean profit per trade
Exposure	Percentage of capital invested
Volatility	Standard deviation of returns

Visualizations commonly include

- Equity curve
 - Drawdown chart
 - Monthly return heatmap
 - Distribution of trade returns
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12. Common Biases & Pitfalls

A reliable backtest must avoid misleading assumptions.

Look-Ahead Bias

Using future information when making historical decisions.

Example: Calculating today's signal using tomorrow's closing price.

Survivorship Bias

Testing only assets that still exist today while ignoring delisted securities.

This can overestimate historical performance.

Data Snooping

Testing many variations until one appears profitable by chance.

Mitigation

- Reserve an out-of-sample dataset.
- Limit repeated optimization.

Overfitting

A strategy that perfectly fits historical data but fails on unseen data.

Indicators

- Excellent in-sample results
 - Poor out-of-sample performance
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13. Walk-Forward Analysis

Walk-forward analysis evaluates a strategy over multiple rolling periods.

Example

Train: 2018–2020

Test : 2021

Train: 2019–2021

Test : 2022

Train: 2020–2022

Test : 2023

Benefits

- Mimics how models are updated in production.
 - Tests adaptability across changing market conditions.
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14. Paper Trading

Paper trading executes the strategy in live markets without risking real capital.

Benefits

- Validate live data feeds.
- Test broker connectivity.
- Measure latency.
- Observe behavior under current market conditions.

It acts as the final validation step before production deployment.

15. Production Readiness Checklist

Before deploying a strategy:

- Historical backtests completed
 - Out-of-sample testing performed
 - Walk-forward validation completed
 - Transaction costs included
 - Slippage modeled
 - Risk limits defined
 - Logging implemented
 - Monitoring dashboards available
 - Alerting configured
 - Rollback procedures documented
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16. Best Practices

- Keep research code separate from production code.
 - Use reproducible datasets and fixed random seeds where appropriate.
 - Version strategies, parameters, and datasets.
 - Record every experiment.
 - Validate assumptions continuously.
 - Benchmark against simple baseline strategies.
 - Automate backtesting within the CI/CD pipeline where possible.
 - Never rely on a single performance metric.
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Summary

Backtesting is the bridge between research and production. A well-designed framework provides realistic execution simulation, robust performance analysis, and safeguards against common statistical pitfalls. Combined with paper trading and ongoing monitoring, backtesting enables developers to evaluate strategies systematically and improve confidence before live deployment.

References

Books

- Ernest P. Chan – *Algorithmic Trading*
- Ernest P. Chan – *Quantitative Trading*
- Marcos López de Prado – *Advances in Financial Machine Learning*
- Yves Hilpisch – *Python for Algorithmic Trading*

Academic Papers

- Bailey et al. – *The Probability of Backtest Overfitting*
- López de Prado – *Advances in Financial Machine Learning* (chapters on validation and backtesting)
- *Qlib: An AI-oriented Quantitative Investment Platform*
- *FinRL: A Deep Reinforcement Learning Library for Automated Stock Trading*

Open Source Frameworks

- Backtrader
- Zipline
- vectorbt
- QuantConnect Lean
- Microsoft Qlib